

Business understanding report

Employee attrition analysis

Analyst lab africa - data science internship (Week 1)

Introduction

When an employee leaves their employer for good, it is called attrition. Some level of employee attrition is normal within most companies; however, when too many employees are leaving, it creates issues for employers. Employers have lost skilled and experienced employees, they have additional costs associated with hiring and training new employees and they often see reduced employee productivity. As stated in the project information, ABC Manufacturing Ltd has chosen to learn why employees are leaving before investing in predictive machine learning models.

Business problem

According to the assignment, ABC Manufacturing Ltd has observed that employees are leaving the company and wishes to determine if there were factors or reasons as to why these employees left. In order to do so instead of simply assuming, ABC Manufacturing Ltd plans to utilize data to help them understand employee behavior and assist them in making informed business decisions. Utilizing this type of analysis will allow ABC Manufacturing Ltd to improve employee retention and establish a positive work environment.

Why does employee attrition matter?

Attrition has a negative effect upon both the company employing the individuals and the employees themselves. Employees with years of service are considered a great asset by companies, since they typically have acquired a significant amount of skills and knowledge. Since experienced employees possess this wealth of knowledge and skills, hiring replacements requires spending money and resources. When employees continuously leave a company, it reduces the overall quality of the team, decreases employee morale, and ultimately delays operational processes. By understanding why employees are leaving a company, employers can take preventative measures to ensure the problem does not continue.

How does data science help support HR departments?

Utilizing data science to analyze employee data assists HR departments in making factual-based decisions as opposed to subjective-based decisions. Through analyzing employee data, organizations can discover patterns and trends that may indicate potential correlations with employee attrition. An example would include discovering through data analysis that employees that frequently work overtime, earn lower wages, or work in particular departments are more likely to leave their employment. Based upon such discoveries, HR departments can formulate improved policies to increase

employee satisfaction and reduce turnover among the workforce. Ultimately, the same data used in this project could be utilized to create predictive machine learning models that identify which employees are at greatest risk of becoming former employees. Therefore, the company can implement intervention techniques to retain those employees.

Project Objectives

This project aims to investigate the employee dataset and respond to important business questions regarding employee attrition. Specifically, this study aims to:

- Understand ABC Manufacturing Ltd's workforce.
- Identify departments experiencing the greatest levels of employee attrition.
- Determine whether age impacts employee attrition.
- Analyze the relationship between monthly earnings and employee retention.
- Explore whether overtime affects employee attrition.
- Determine what positions exhibit the greatest rates of employee turnover.
- Develop variables to support future predictive analytics and machine learning studies.

What is expected from me in this project?

Through the results of this analysis, ABC Manufacturing Ltd will gain greater insight into its workforce and the probable causes for employees departing from their position. Such insights will enable HR departments to make educated decisions concerning employee well-being, workload, compensation, and retention initiatives. Moreover, the data collected for this analysis will serve as a solid foundation for developing future predictive analytical and machine learning research opportunities.(I can't wait for this task)

Conclusion

Attrition is one of many critical business concerns affecting productivity, expenses, and long term success of an organization. By investigating employee data and determining relevant characteristics contributing to attrition, ABC Manufacturing Ltd will be able to make informed decisions relative to improving employee retention. This project

represents the initial stage in utilizing data science to address actual business issues and inform better HR decision-making practices.